

4. The table shows the composition of four particles, **W**, **X**, **Y** and **Z**.

Particle	Number of protons	Number of electrons	Number of neutrons
W	12	12	12
X	12	12	14
Y	12	10	12
Z	11	11	12

- (a) Use information from the table to answer parts (i)-(iii). [1]
- (i) State how the information shows that particle **X** is an atom. [1]
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- (ii) State how the information shows that particle **Y** is a positive ion. [1]
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- (iii) State how the information shows that particles **W** and **X** are isotopes of the same element. [1]
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- (b) (i) Give the electronic structure of element **Z**. [1]
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- (ii) State the group and period number of element **Z** in the Periodic Table. [1]
- Group
- Period

